

Biomass Prediction from Images by Convolutional Network

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Abstract— A convolutional neural network (CNN) was developed to predict pasture biomass metrics directly from JPEG images. This was conducted using data from the Kaggle competition *CSIRO – Image2Biomass Prediction*. The dataset consists of labeled pasture images for training and one unlabeled image for testing and contest submission, with quantitative ground-truth biomass measurements provided for the training set. To evaluate model performance, the labeled data was split into training and validation subsets. A CNN was developed using a transfer learning approach based on a pre-trained convolutional architecture. Training was conducted in stages, incorporating callbacks to improve convergence and prevent overfitting, followed by fine-tuning of selected network layers. The final model was designed to predict five continuous pasture biomass components from the JPEG imagery. Despite these efforts, the model did not achieve meaningful predictive performance as the R^2 values were near zero. This was likely due to the limited size of the training dataset and the high variability inherent in pasture composition and environmental conditions.

I. INTRODUCTION (HEADING I)

A. Background

With the increasing impacts of climate change, pollution, and loss of suitable grazing land, effective pasture management is becoming increasingly important. Forage—the edible, nutritious plant matter that constitutes a major part of many animals’ diets—makes up approximately 80% of goat and sheep nutrition, 73% of beef cattle nutrition, and 51% of a horse’s diet. (Washington State University Extension) Effective pasture grazing is challenged by variability in forage quality and quantity. As noted by Washington State University, “One major difficulty with pasture grazing of animals is inconsistency of forage quality and quantity” (Washington State University). Standard methods for determining pasture biomass include destructive sampling techniques such as clip-and-weigh and indirect measurement tools like plate and capacitance meters. (CSIRO). While clip-and-weigh is highly accurate, it is labor-intensive and slow, making it impractical at large scales. Using meters can produce faster results but are often less accurate and sensitive to environmental conditions. Accurate assessment of pasture biomass is critical for sustainable grazing management. Without precise knowledge of available forage quantity and composition, farmers risk

overgrazing, land degradation, or inadequate nutrition for livestock. Reliable biomass estimates enable better decision-making, supporting both productivity and ecosystem health (CSIRO).

B. The goal of the competition

The objective of the *CSIRO – Image2Biomass Prediction* competition is to estimate five key pasture biomass components directly from images:

- Dry green vegetation (excluding clover)
- Dry dead material
- Dry clover biomass
- Green dry matter (GDM)
- Total dry biomass

Accurate estimation of these components enables farmers and land managers to optimize grazing strategies, improve productivity, and protect pasture ecosystems. The competition frames this task as a supervised learning problem, using labeled image data as input and continuous biomass measurements as output.

II. LITERATURE

A. Pasture land usage

There are four major types of grazing systems. Continuous grazing, Deferred Rotational grazing, Rest Rotational grazing, and Management-intensive grazing. Continuous grazing is when the herd stays on the same pasture (Marshall). This is a lowcost solution at first, but the land will be at risk of being overgrazed and depleted. Deferred Rotational grazing is when certain areas are allowed to rest. This prevents land from being completely depleted. Rest rotational grazing divides pasture into sections and allows each section to rest for an entire season. This conserves the environment, land, and water. Rotating land requires significantly more labor and infrastructure resources than a continuous grazing system. However, the long-term benefits are usually worth it. Management-intensive grazing involves frequent movements of livestock into small paddocks. This optimizes forage utilization and sustainability. Utilization of forage can be

increased by 20 to 30%. However, this can be challenging to implement and requires even more resources than the rotation methods. The goal of this project is to make the high potential of a Management-intensive grazing system more feasible.

B. Traditional methods of Assessing paddocks

Hand-Clipping and Oven Drying has been considered the best way to get an accurate measurement of Total dry biomass in a paddock. This method involves clipping a sample of the area to ground, drying the sample for days in an oven, and then weighing the dry sample. If the sample is sorted, all five key pasture biomass components can be measured. Although accurate, this method damages the sampled area and requires a lot of labor and time. (Voelkel, B., 2011)

A less intrusive way of determining the nutrition of a pasture is by Rising Plate Meter. A plate lowers a pole onto the pasture vegetation and measures height and density. The data collected is then used to calculate Dry Matter and Total standing dry matter. This method is quick and non-destructive, but requires calibration and is prone to overestimating. (Harmony)

Another quick and less labor intensive method to evaluate the land is to use an Active Optical Sensor.

C. Use of AI in Agriculture

The field of agriculture has increasingly adopted artificial intelligence (AI) over the past decade. AI models have been developed to monitor livestock through biometrics, assess plant health, detect pests and weeds, manage fields, and automate sorting and grading of crops. (Menezes et al., 2025). These advancements aid an industry plagued with labor-intensive tasks and labor shortages.

III. DATA ACQUISITION

This dataset included test.csv, training.csv, sample_submission.csv, training images, and a test image. A training csv with labels for each image with image paths was provided.

A. Training Images

The competition dataset provides a collection of RGB images used to train the convolutional neural network. Each image represents a paddock with varying types and densities of vegetation cover. These images capture substantial visual variability due to differences in plant species composition, ground cover, lighting conditions, and growth stages.

The convolutional neural network learns directly from these images, extracting spatial and texture-based features that may correlate with pasture biomass composition. Example training images are shown below to illustrate the diversity of field conditions present in the dataset.

Fig 1



Fig 2



Fig 3



B. Training csv file and labels

The dataset also includes a training.csv file that provides metadata and target information associated with each image. Each row in the CSV contains the following attributes:

- sample_id: Unique identifier for each sample
- image_path: File path to the corresponding image
- Sampling_Date: Date the image was collected
- State: Australian state where image was taken
- Species: Dominant vegetation species
- Pre_GSHH_NDVI: Vegetation index measured prior to sampling
- Height_Ave_cm: Average vegetation height in centimeters
- target_name: Name of the biomass attribute being measured
- target: The quantitative biomass value associated with the target

The project objective is to predict the CSV biomass values for each target category directly from the image data. The dataset includes five biomass targets:

- Dry_Clover_g
- Dry_Dead_g
- Dry_Green_g
- Dry_Total_g
- GDM_g (Green Dry Matter)

Because each image appears multiple times in the CSV (once for each target), the data was reorganized so that each image corresponds to a single training instance with five associated target values. This restructuring enabled the model to perform multi-output regression, predicting all biomass categories simultaneously from a single image.

C. Test CSV and test image

The test file only provided the image path for inputs and did not include the Height_Ave_cm or Pre_GSHH_NDVI. Height_Ave_cm is critical information when predicting biomass targets. Since this information was provided in the training CSV. The model was trained to learn these values.

III. METHODOLOGY

A convolutional neural network (CNN) was used to model and predict pasture composition from JPEG images of paddocks. The task was formulated as a multi-output regression problem, where the model predicts five biomass-related categories for each image. The methodology consisted of data restructuring, dataset construction, transfer learning using a pre-trained network, and fine-tuning to improve model performance.

A. Data Processing

No explicit data cleaning was required because the image dataset and corresponding CSV files were generated through a controlled data collection process. The images were used *as-is*, allowing the convolutional neural network to learn directly from the raw visual information without manual feature engineering.

However, the original training CSV was not organized in a format suitable for supervised learning with CNNs. Each row in the CSV corresponded to a single biomass measurement, rather than a single image. To address this, the data was reorganized into a wide-format DataFrame, where each row represents one image and the columns correspond to the five target biomass categories. This restructuring ensured that all target values associated with an image could be predicted simultaneously by the model. The data was split into training and valuation data with 80:20 split in order to evaluate model performance.

During dataset creation, images were:

- Loaded from disk using their file paths,
- Decoded into RGB format,
- Resized to a fixed resolution of 224×224 pixels, ensuring consistent tensor shapes,

- Converted to floating-point format and normalized prior to model input.

The final dataset was constructed using the TensorFlow tf.data API, enabling efficient batching, prefetching, and parallel loading during training and evaluation.

B. Dataset Structure and Input Representation

Each training and validation instance consisted of:

- An input tensor of shape **(224, 224, 3)** representing the RGB image, and
- A target vector of length five representing the biomass categories.

Batching was performed with a fixed batch size, while allowing the final batch to contain fewer samples to prevent tensor shape errors during prediction. Shuffling was applied only to the training dataset to improve generalization. This pipeline ensured that the CNN received consistently shaped inputs and that the dataset could be used seamlessly for training, validation, and inference.

C. Use of a Pre-trained Network

The model architecture was based on EfficientNetB0, a convolutional neural network pre-trained on the ImageNet dataset. Transfer learning was employed to leverage the rich feature representations learned from large-scale natural image data.

The pre-trained EfficientNet backbone was used without its classification head (include_top=False) and served as a fixed feature extractor during initial training. A custom regression head consisting of:

- Global average pooling,
- Fully connected layers with ReLU activation,
- Dropout regularization,
- A linear output layer with five neurons,

was added to produce continuous-valued predictions for the five biomass targets.

Freezing the pre-trained layers during early training stabilized optimization and reduced the risk of overfitting, particularly given the limited size of the dataset.

D. Fine-Tuning Strategy

After training the newly added regression layers, fine-tuning was performed to improve predictive performance. Fine-tuning involved selectively unfreezing the EfficientNet backbone and allowing its weights to adapt to the pasture imagery domain.

To prevent destabilizing the pre-trained features:

- A lower learning rate was used during fine-tuning,
- Batch normalization layers were optionally kept frozen to maintain stable feature statistics,

- Training was continued for a limited number of epochs.

Model training employed callbacks including:

- Early stopping, to halt training when validation performance stopped improving, and
- ReduceLROnPlateau, to decrease the learning rate when validation loss plateaued.

This staged training approach allowed the model to benefit from both pre-trained representations and domain-specific adaptation while minimizing overfitting and training instability.

IV. RESULTS

Model performance was evaluated using mean squared error (MSE)(Fig 6), Root Mean Squared Error (RMSE)(Fig 4), and the coefficient of determination (R^2)(Fig 5) for each of the five predicted biomass variables. These metrics were selected to quantify the average magnitude of prediction error and to evaluate how much of the variability in the observed biomass values was explained by the model.

The MSE and RMSE values both suggest that the predicted values often strayed significantly from the ground truth. Both also imply that values for the Dry_Clover_g and the Dry Dead_g were better predicted than the other measurements.

The R^2 values for the Biomass are near zero. These R^2 values suggest that the model struggled to capture meaningful relationships between image features and the target biomass measurements.

The final evaluation metrics indicate that the model does not yet achieve strong predictive performance. While the model produced numerically valid outputs for all five targets, the predicted values showed limited agreement with expected biomass distributions, indicating insufficient generalization. Despite utilizing a pre-trained model and fine-tuning, the available data and model capacity were not sufficient to reliably predict quantitative pasture composition.

Fig 4 : RMSE for the 5 biomass targets

Dry_Clover_g	RMSE: 13.405
Dry_Dead_g	RMSE: 12.103
Dry_Green_g	RMSE: 21.921
Dry_Total_g	RMSE: 24.938
GDM_g	RMSE: 22.113

Fig 5: R^2 for the 5 biomass targets

Dry_Clover_g	R^2 : -0.008
Dry_Dead_g	R^2 : -0.013
Dry_Green_g	R^2 : -0.022
Dry_Total_g	R^2 : -0.068
GDM_g	R^2 : -0.017

Fig 6: MAE for the 5 biomass targets

Dry_Clover_g	MAE: 8.891
Dry_Dead_g	MAE: 9.098
Dry_Green_g	MAE: 15.695
Dry_Total_g	MAE: 19.314
GDM_g	MAE: 15.864

V. DISCUSSION

This task proved to be particularly challenging. Predicting multiple continuous-valued pasture biomass components from JPEG imagery alone is a complex problem that typically requires large datasets and carefully tuned models. In this study, several factors limited model performance. First, the dataset contained only 357 labeled images, which is relatively small for a multi-output regression task and constrained the model's ability to generalize. Second, the visual differences among biomass categories are often subtle and may not be fully captured by RGB imagery, particularly for distinguishing dry green, dry dead, and clover components. Third, more computational power might be needed in order to run more epochs while fitting the model. Fourth effectively fine-tuning a pre-trained model might require more time. Despite hyperparameter tuning, gradual layer unfreezing, and the use of callbacks such as early stopping and learning rate reduction, the model was not able to make reliable predictions.

VI. CONCLUSION

This project demonstrated the difficulty of predicting multiple quantitative biomass variables directly from labeled pasture images. While transfer learning and fine-tuning improved model behavior compared to training from scratch, the final results indicate that the model is not yet capable of making reliable predictions.

Estimating five continuous-valued biomass metrics from a relatively small image dataset is inherently challenging. The variability in plant species composition, lighting conditions, and ground cover could have contributed to these challenges. With only 357 images available, the dataset may be insufficient to support accurate multi-target regression. Future work could explore simplifying the task, such as reframing it as a classification problem (e.g., whether land should be grazed or rested), incorporating additional data sources (e.g., multispectral imagery or environmental variables), or expanding the dataset size. These directions may lead to more robust and interpretable results.

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